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## Devices and methods for nitrosylation of blood


### Abstract

Devices and methods are provided herein for exchange between a first agent and a second agent. The device includes a first chamber having a first inlet and a first outlet with a first flow passageway extending between the first inlet and the first outlet for flowing a first agent through the first chamber. The device also includes a second chamber having a second inlet and a second outlet with a second flow passageway extending between the second inlet and the second outlet for flowing a second agent through the second chamber and a membrane positioned between the first flow passageway and the second flow passageway to allow exchange through the membrane of the first agent with the second agent. Methods of nitrosylating blood and methods of manufacturing the device are also provided.



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## Background/Summary

CLAIM OF PRIORITY (1) This application is the national phase application of PCT/US2018/036413, filed on Jun. 7, 2018, which claims priority from U.S. Ser. No. 62/516,985 filed Jun. 8, 2017, both of which are is incorporated herein by reference in their entirety.

BACKGROUND
1. Technical Field
(1) This application relates to nitrosylation of blood, and in particular to devices and methods for nitrosylation of blood for transfusion.
2. Background Information
(2) It is desirable to nitrosylate blood prior to transfusion to restore nitric oxide (NO) bioactivity. Previously, pediatric oxygenators have been used to deliver ethyl nitrite (ENO) gas to attempt to restore NO bioactivity. However, nitrosylation is complicated by time-dependent variability during

transfusion with SNO levels varying from almost zero to physiological range. In addition, variability in SNO levels has occurred during controlled flow rates, which can vary dramatically during transfusion.
(3) Provided herein are a device for nitrosylating blood ex vivo with ENO, in gas or liquid form, and a method for using the device for the nitrosylation of blood ex vivo with ENO, in gas or liquid form. Methods of manufacturing the device are also provided.
BRIEF SUMMARY
(4) Devices and methods are provided herein for exchange between a first agent and a second agent.
(5) In one aspect, a device for exchange between a first agent and a second agent is provided. The device includes a first chamber having a first inlet and a first outlet with a first flow passageway extending between the first inlet and the first outlet for flowing a first agent through the first chamber. The device also includes a second chamber having a second inlet and a second outlet with a second flow passageway extending between the second inlet and the second outlet for flowing a second agent through the second chamber and a membrane positioned between the first flow passageway and the second flow passageway to allow exchange through the membrane of the first agent with the second agent.
(6) In an embodiment, the first chamber and the second chamber each comprise a plurality of transport channels.
(7) In an embodiment, the first chamber and the second chamber each comprises a first passageway and a second passageway.
(8) In an embodiment, the first passageway and the second passageway each comprise structural members formed in the first and second passageways.
(9) In an embodiment, the first chamber and the second chamber each comprise flow distribution members.
(10) In an embodiment, the flow distribution members are positioned upstream from a plurality of transport channels in each of the first chamber and the second chamber.
(11) In an embodiment, the device comprises a single module device.
(12) In an embodiment, the device comprises a multiple module device comprising a plurality of first chambers and a plurality of second chambers.
(13) In an embodiment, a wall of the first chamber or a wall of the second chamber forms the membrane for each module.
(14) In an embodiment, the first chamber and the second chamber comprise poly(dimethyl siloxane) (PDMS).
(15) In an embodiment, the membrane is permeable to a nitrosylating agent.
(16) In an embodiment, the nitrosylating agent comprises ethyl nitrite.
(17) In an embodiment, the device comprises a flow rate therethrough of about 10-1,200 μl/min. In an embodiment, the device comprises a flow rate therethrough of about 100 to about 1,100 μl/min, about 200 to about 1,000 μl/min, about 300 to about 900 μl/min, about 400 to about 800 μl/min, or about 500 to about 700 μl/min. In an embodiment, the device comprises a flow rate therethrough of about 1 to about 100 μl/min, about 100 to about 200 μl/min, about 200 to about 300 μl/min, about 300 to about 400 μl/min, about 400 to about 500 μl/min, about 500 to about 600 μl/min, about 600 to about 700 μl/min, about 700 to about 800 μl/min, about 800 to about 900 μl/min, about 900 to about 1,000 μl/min, about 1,000 to about 1,100 μl/min, or about 1,100 to about 1,200 μl/min.
(18) In another aspect, a method of nitrosylating blood is provided. The method includes flowing blood through a first chamber of a device where the first chamber of the device is separated from a second chamber of the device by a membrane and flowing a nitrosylating agent through the second chamber of the device so that the blood is nitrosylated by the nitrosylating agent by exchange through the membrane.
(19) In an embodiment, the nitrosylating agent is selected from the group consisting of ethyl nitrite,

amyl nitrite, butyl nitrite, isobutyl nitrite, and tert-butyl nitrite. In an embodiment, the nitrosulating agent is ethyl nitrite. In an embodiment, the nitrosulating agent is amyl nitrite. In an embodiment, the nitrosulating agent is butyl nitrite. In an embodiment, the nitrosulating agent is isobutyl nitrite. In an embodiment, the nitrosulating agent is tert-butyl nitrite.
(20) In an embodiment, the nitrosylating agent comprises ethyl nitrite (ENO).
(21) In an embodiment, the method comprises flowing liquid ENO or ENO gas through the second chamber.
(22) In an embodiment, the method comprises flowing the blood through the device at a flow rate of about 10-1,200 μl/min. In an embodiment, the device comprises a flow rate therethrough of about 100 to about 1,100 μl/min, about 200 to about 1,000 μl/min, about 300 to about 900 μl/min, about 400 to about 800 μl/min, or about 500 to about 700 μl/min. In an embodiment, the device comprises a flow rate therethrough of about 1 to about 100 μl/min, about 100 to about 200 μl/min, about 200 to about 300 μl/min, about 300 to about 400 μl/min, about 400 to about 500 μl/min, about 500 to about 600 μl/min, about 600 to about 700 μl/min, about 700 to about 800 μl/min, about 800 to about 900 μl/min, about 900 to about 1,000 μl/min, about 1,000 to about 1,100 μl/min, or about 1,100 to about 1,200 μl/min.
(23) In an embodiment, the method comprises nitrosylating the blood to a final range of about 1-100 SNO/1000 Hb levels. In an embodiment, the method comprises nitrosylating the blood to a final range of about 10 to about 90 SNO/1000 Hb levels, about 20 to about 80 SNO/1000 Hb levels, 30 to about 70 SNO/1000 Hb levels, about 40 to about 60 SNO/1000 Hb levels, or about 50 SNO/1000 Hb levels. In an embodiment, the method comprises nitrosylating the blood to a final range of about 1 to about 90 SNO/1000 Hb levels, about 1 to about 80 SNO/1000 Hb levels, about 1 to about 70 SNO/1000 Hb levels, about 1 to about 60 SNO/1000 Hb levels, about 1 to about 50 SNO/1000 Hb levels, about 1 to about 40 SNO/1000 Hb levels, about 1 to about 30 SNO/1000 Hb levels, about 1 to about 20 SNO/1000 Hb levels, about 1 to about 10 SNO/1000 Hb levels.
(24) In yet another aspect, a method of manufacturing a device for exchange between a first agent and a second agent is provided. The method includes providing a template, filling a cavity of the template with a polymer, and curing the polymer. The method also includes separating the cured polymer from the template to form a first chamber of a device and connecting the first chamber to a second chamber so that a membrane separates the first chamber from the second chamber.
(25) In an embodiment, the polymer comprises poly(dimethyl siloxane) (PDMS).
(26) In an embodiment, the method comprises providing a template having a plurality of structural members, a plurality of flow modification members and a plurality of transport chambers.
(27) In an embodiment, the method comprises assembling the first chamber and the second chamber to form a single module device.
(28) In an embodiment, the method comprises connecting tubing to an inlet and an outlet of the first chamber for flowing blood through the first chamber.
(29) In an embodiment, the method comprises assembling a plurality of first chambers and second chambers together, each first chamber and second chamber having the membrane separating the first chamber from the second chamber.
(30) In an embodiment, the method comprises providing a wall of the first chamber or the second chamber to form the membrane separating the first chamber from the second chamber.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1**A illustrates an embodiment of a module of a device for nitrosylation.
(2) FIG. **1**B illustrates a partial sectional view of the device shown in FIG. **1**A.
(3) FIG. **2** illustrates an embodiment of two portions for forming the device shown in FIG. **1**A

showing alternating patterns for the portions.
(4) FIG. **3** illustrates partial view of an embodiment of a template for forming the portions of the device.
(5) FIG. **4**A illustrates a simulation showing uniform flow distribution in an embodiment the device.
(6) FIG. **4**B illustrates a simulation showing uniform flow distribution in the individual transport channels of an embodiment of the device.
(7) FIG. **5**A illustrates a partial cross-sectional view of an embodiment of the device having multiple modules.
(8) FIG. **5**B illustrates a top view of an embodiment of a device having a frame.
(9) FIG. **6** illustrates an embodiment of a device that can be made including 50 modules assembled together.
(10) FIG. **7** illustrates an example showing NO exit levels versus flow rate.
(11) FIGS. **8**A and **8**B illustrate embodiments of devices having different amounts of nitrosylating agent being exchanged with blood. (**8**A: low, **8**B: high).
(12) FIG. **9** illustrates an example of SNO/1000 Hb levels in two outlet samples versus an inlet sample.
(13) FIG. **10** illustrates an example of a photolithography process that may be used to form a template.
(14) FIG. **11** illustrates an example of a soft lithography process that may be used with the template to form a chamber of the device.

DETAILED DESCRIPTION

(15) The embodiments disclosed below are not intended to be exhaustive or to limit the scope of the disclosure to the precise form in the following description. Rather, the embodiments are chosen and described as examples so that others skilled in the art may utilize its teachings.
(16) As used herein, the term “about” means approximately, in the region of, roughly, or around. When the term “about” is used in conjunction with a numerical range, it modifies that range by extending the boundaries above and below the numerical values set forth. In general, the term “about” is used herein to modify a numerical value above and below the stated value by a variance of 10%.
(17) As used herein, the phrases “to cure the polymer,” “curing the polymer” or “cured polymer” refers to the toughening or hardening of a polymer material by cross-linking of polymer chains, e.g., brought about by electron beams, ultraviolet radiation, heat, or chemical additives.
(18) A device **100** for modifying blood ex vivo is shown in FIGS. **1**A and **2**. The device **100** includes a first inlet **102** and a first outlet **104** connected to a first chamber **106** having a first flow passageway **107** (shown in FIG. **2**), for flowing a first agent **108** through the device **100**. The first flow passageway **107** may extend from the first inlet **102** to the first outlet **104** through the device **100**. In some embodiments, the first agent **108** may be blood. The device **100** may also include a second inlet **105** and a second outlet **110** connected to a second chamber **112** having a second flow passageway **109** for a second agent **111** in the device **100**. The second flow passageway **109** may extend from the second inlet **105** to the second outlet **110** through the device **100**. The second agent **111** may be in gas or liquid form. In some embodiments, the second agent may be a nitrosylating agent. By way of non-limiting example, the nitrosylating agent may be ethyl nitrite, amyl nitrite, butyl nitrite, isobutyl nitrite, tert-butyl nitrite and combinations thereof. In an embodiment, the nitrosulating agent is ethyl nitrite. In an embodiment, the nitrosulating agent is amyl nitrite. In an embodiment, the nitrosulating agent is butyl nitrite. In an embodiment, the nitrosulating agent is isobutyl nitrite. In an embodiment, the nitrosulating agent is tert-butyl nitrite. In an embodiment, the nitrosylating agent comprises ethyl nitrite (ENO). In some embodiments, the second agent **111** may be oxygen or any other agent **111** desirable to infuse into the first agent **108**. The device **100** in FIG. **1**A shows the first and second chambers **106**, **112** assembled together. The first and second

chambers **106**, **112** mirror each other and form one module **116** of the device **100**. When one single module **116** is used, a membrane **120** may be positioned between the first and second chamber **106**, **112** as shown in FIG. **1**B. The membrane **120** is permeable to allow exchange between the first agent **108** and the second agent **111**. For example, the membrane **120** allows for nitrosylation of blood when blood is flowed through the first chamber **106** and a nitrosylating agent **111** is in the second chamber **112**. The membrane **120** may be hydrophobic porous or non-porous polymer material. When multiple modules **116** are used together, a wall **122** of the first chamber **106** or the wall **122** of the second chamber **112** of one module **116** may form the membrane **120** of the adjacent module **116** having similar permeability as described in more detail below.

(19) FIG. **2** illustrates the two portions of the device **100** that are mated together to form the module **116** shown in FIG. **1**A. Two alternating patterns may be used to form the mirror images that join to form the chambers **106**, **112** for the first agent **108** and the second agent **111**. The alternating inlet **102** and outlet **104** allow for a single inlet **102** and a single outlet **104** even when multiple modules **116** are stacked together. In some embodiments, multiple inlets **102** and outlets **104** may be used. Similarly, a single second inlet **105** and a single second outlet **110** or multiple inlets **105** and outlets **110** may be used for the second chambers **112** when multiple modules **116** are assembled together. Tubing may be connected to the inlets and outlets for flowing the substances into and out of the modules **116**. The chambers **106**, **112** are described herein with reference to chamber **106**. The chamber **112** is the mirror image of the chamber **106** and includes the same features as the chamber **106** described herein. As shown in FIGS. **1**A and **2**, the chamber **106** includes a first passageway **124** connected to the inlet **102**. The first passageway **124** is connected to a central portion **126** that is connected to a second passageway **128** connected to the outlet **104**.

(20) FIG. **3** illustrates an example of a portion of a template **130** that may be used to form a pattern **132** in the walls **122** that form the chambers **106**, **112** of the device **100** shown in FIG. **1**A. FIG. **3** corresponds to the pattern **132** that is shown in the box in FIG. **1**A. The walls **122** may be made from a polymer using the template **130** to mold the pattern **132** in the walls **122** of the device **100** to create the chambers **106**, **112**. As shown in FIG. **3**, the template **130** includes several features for support and flow control of the first and/or second agents **108**, **111** through the chambers **106**, **112**. The template **130** is used to form a plurality of support structures **140** (also indicated by *). As shown in FIGS. **1**A and **2**, the support structures **140** may be formed in the first passageway **124** and the second passageway **128**. In some embodiments, the support structures **140** may extend into the central portion **126** of the chambers **106**, **112**. In some embodiments, the support structures **140** may be cylindrical, however, other shapes may also be used. By way of non-limiting example, support structures may be cylindrical, oval, triangular, polygonal, etc. The cylindrical structures **140** may be about 500 μm, for example for use with a 50 μm thick membrane **120** that separates the first agent **108** from the second agent **111**. In some embodiments, the size range of the cylindrical structures may be about 100-1000 μm.

(21) The template **130** may also be used to form flow distribution members **144** that may be positioned in the central portion **126** of the chambers **106**, **112**. As shown in FIG. **3**, the flow distribution members **144** may be of different sizes and shapes. The flow distribution members **144** enable uniform distribution of flow in individual transport channels **148** in the chambers **106**, **112**. In some embodiments, the flow distribution members **144** may be tapered inward and/or outward to facilitate smooth flow through the chambers **106**, **112**. The flow distribution members **144** may be positioned on the upstream and/or downstream side of the individual transport channels **148**. In some embodiments, the individual transport channels **148** may be about 100 μm (width)×100 μm (depth)×10 mm (length). In other embodiments, the transport channels **148** may be 50 μm (width)×50 μm (depth) and may be varied down to 10 μm. In some embodiments, the length of the individual transport channels may be about 10 mm to 100 mm. Each chamber **106**, **112** may include 2.sup.n channels **148**, where n can range from 6 to 10. In some embodiments, the device **100** may include 32 or 64 transport channels **148**. The size and number of the support structures

**140**, the flow distribution members **144** and the individual transport channels **148** will depend on the flow rate, the volume and the agents. In a single module **116** having 64 individual transport channels and dimensions 100 μm (width)×100 μm (depth)×10 mm (length), the area for interface between the first and the second agents is about 4.25 cm.sup.2.

(22) FIG. **4**A shows a simulation showing uniform flow distribution in the device **100** and FIG. **4**B shows a simulation showing uniform flow distribution in the individual transport channels **148**. The design of the chambers **106**, **112** and the individual transport channels **148** allow for even flow and exchange between the agent **108** in the first chamber **106** and the second agent **111** in the second chamber **112** of the device **100** shown in FIG. **1**.

(23) FIGS. **5**A, **5**B and **6** illustrate the device **100** having a plurality of modules **116** assembled together. FIG. **5**A is a cross-section through the individual transport channels **148** of the device **100** showing alternating first agent **108** and second agent **111** in the transport channels **148** with the wall **122** forming the membrane **120** between the first agent **108** and the second agent **111** where multiple modules **116** are stacked together. The wall **122** may be part of the first chamber **106** and/or the second chamber **112** when multiple modules **116** are stacked together so that the wall **122** forms the membrane **120**. The wall **122** may be formed to be thin enough for exchange between the first agent **108** and the second agent **111**. FIG. **5**A illustrates seven alternating layers of transport channels **148** representing four modules **116**. FIG. **6** illustrates an example showing that 50 modules **116** can be assembled together to form the device **100**. Other numbers of modules **116** may be assembled together to form the device **100**. The number of modules **116** assembled together may be any number and may be varied depending on the flow rate, the agents and the volume needed. In some embodiments, the device **100** having multiple modules **116** assembled together may include an outer frame **160** for additional support as shown in FIG. **5**B. In some embodiments, the support frame **160** may be made from a metal, such as steel, including stainless steel.

(24) In some embodiments, the overall thickness of each module **116** may be about 200 μm. A 1 mm thick device **100** includes about 5 modules **116** and a 1 cm thick device includes about 50 modules **116**. In a 50 module device having 50 μm (width)×50 μm (depth) channels **148**, the contact area for the first agent **108** and the second agent **111** through the membrane **120** is about 10.sup.3-10.sup.4 cm.sup.2/cm.sup.3. By way of non-limiting example, when the first agent is blood and the second agent is a nitrosylating agent, the overall nitrosylation transport area in the device is about 1000 cm.sup.2/cm.sup.3. The volumes of the chambers **106** and **112** are about the same, in some embodiments the volume may be approximately 1.2 cm.sup.3. In some embodiments, the nitrosylation capability of the device **100** in blood is about 0.37 SNO/1000 Hb/10 cm.sup.2/50 ppm eNO driving force/125 μm polymer, using a 1 ml/min blood flow as a basis for the calculation. In some embodiments, the flow through each module **116** may range from about 10-1,200 μl/min and remain free from leaks of agents from the device **100**. In an embodiment, the device comprises a flow rate therethrough of about 100 to about 1,100 μl/min, about 200 to about 1,000 μl/min, about 300 to about 900 μl/min, about 400 to about 800 μl/min, or about 500 to about 700 μl/min. In an embodiment, the device comprises a flow rate therethrough of about 1 to about 100 μl/min, about 100 to about 200 μl/min, about 200 to about 300 μl/min, about 300 to about 400 μl/min, about 400 to about 500 μl/min, about 500 to about 600 μl/min, about 600 to about 700 μl/min, about 700 to about 800 μl/min, about 800 to about 900 μl/min, about 900 to about 1,000 μl/min, about 1,000 to about 1,100 μl/min, or about 1,100 to about 1,200 μl/min.

(25) The device **100** may nitrosylate blood to a stable physiological range of 1-5 SNO/1000 Hb levels and up to a range of 1-100 SNO/1000 Hb levels. In an embodiment, the method comprises nitrosylating the blood to a final range of about 10 to about 90 SNO/1000 Hb levels, about 20 to about 80 SNO/1000 Hb levels, 30 to about 70 SNO/1000 Hb levels, about 40 to about 60 SNO/1000 Hb levels, or about 50 SNO/1000 Hb levels. In an embodiment, the method comprises nitrosylating the blood to a final range of about 1 to about 90 SNO/1000 Hb levels, about 1 to about 80 SNO/1000 Hb levels, about 1 to about 70 SNO/1000 Hb levels, about 1 to about 60

SNO/1000 Hb levels, about 1 to about 50 SNO/1000 Hb levels, about 1 to about 40 SNO/1000 Hb levels, about 1 to about 30 SNO/1000 Hb levels, about 1 to about 20 SNO/1000 Hb levels, about 1 to about 10 SNO/1000 Hb levels.

(26) The device **100** may be scaled for use in multiple transfusion conditions including: standard (1 unit of blood every 2-4 hours), rapid (1 unit of blood every hour), trauma (1 unit of blood every 10-15 minutes) or massive trauma (1 unit of blood every minute). In some embodiments, the flow rate may be about 500 ml/minute.

(27) FIG. **7** illustrates a graph showing control of nitrosylation of blood using the module **116** described above. The amount of NO in the blood exiting the device **100** may be modified by changing the flow rate through the device **100**.

(28) FIGS. **8**A and **8**B illustrate differences in the amount of the nitrosylating agent supplied to the device **100** in the second chamber **112** and exchanged with the blood in the first chamber **106**. FIG. **8**A shows a lower amount of nitrosylating agent being exchanged with blood flowing through the device **100** than FIG. **8**B. FIG. **8**B shows oxidation of the blood (demonstrated by the lighter color of blood flowing past the channels and out of the device). By way of non-limiting example, a device **100** having the membrane **120** with a thickness of about 50 μm between the first and second agents **108**, **111** has a specific permeability value of about 8×10.sup.−5 cm.sup.2/min for the nitrosylating agent in a membrane **120** made from poly(dimethyl siloxane). The nitrosylation of the blood can be controlled using three design parameters of the device **100** including membrane thickness, overall flow rate of blood and nitrosylation agent levels. The device **100** is also compatible with oxygen transport.

(29) FIG. **9** shows an example of renitrosylation of 7-day old blood samples using 50 ppm ENO as the second agent **111** in the second chamber **112** in a single module device **100** with a flow rate of 1 ml/min of blood in the first chamber **106** and a single pass through the device **100**.

(30) Methods for making the device **100** are also described herein. The modules **116** of the device may be made from different polymeric materials that can be used with the template **130** to form the pattern **132** for each chamber **106**, **112** of the device **100**. In some embodiments, different molecular weights of a polymer forming the chambers or membranes may be used to control nitrosylation rates. In some embodiments, poly(dimethyl siloxane) (PDMS) may be used to form the modules **116**. Other polymers that may be used include UV curable vinyl ether polymers, poly(methyl methacrylate), but are not limited to these polymers.

(31) In some embodiments, a carrier frame may be provided in which to build the modules described above. The frame may be made of stainless steel that is fabricated using photochemical milling. Each frame can hold multiple modules **116**. In some embodiments, a frame may be milled to hold hundreds or thousands of modules. The frame may be sized and shaped to hold a plurality of templates for forming multiple chambers to make the modules. The templates may be registered within the frame using guide pins. In some embodiments, the templates may be manufactured using photolithography to produce one or more silicon wafer templates. Soft lithography may then be used to produce each layer/chamber of the device from the template.

(32) FIG. **10** illustrates an example of a photolithography process that may be used to form the template. Other processes known to one skilled in the art may also be used to form the template. As shown in FIG. **10**, the template may be manufactured by covering a silicon wafer (e.g., University Wafer, Boston, MA) with photoresist (e.g., Su-850 photoresist, Micorchem, Newton, MA) by spin coating. The photoresist coated wafer is then soft baked. A mask is layered over the photoresist coated silicon wafer and the photoresist and mask are exposed to UV light. A post-UV exposure bake is performed and the wafer is developed (Su-8 developer, Microchem, Piranha Solution (3:1 H2SO4:H202 (30%)) with photoresist features that may be used for the template for forming the chambers of the device.

(33) FIG. **11** illustrates an example of a soft lithography process for forming a layer/chamber of the device using the template. Other processes known to one skilled in the art may also be used to form

the chambers. Each cavity is filled with an uncured polymer, such as PDMS (e.g., Sylgard 184 base and curing agent, Dow Corning, MI). The template/polymer assembly is heated to cure the polymer. The cured polymer and template are separated (e.g., silanizing agent, tridecafluoro-1,1,2,2-tetrahydrooctyl trichlorosilane, United Chem, Horsham, PA) so that the cured polymer forms a layer/chamber of the device. In some embodiments, the template may be removed from the frame leaving the cured polymer in the frame. In some embodiments, the cured polymer may be removed from the frame. In a single module, a membrane may be added between the layers/chambers and connected using oxygen plasma bonding. The holes at the inlets and outlets may be punched out and tubing connected to the inlets and outlets. In a multimodular device, the alternating patterns may be layered on top of each other to form a plurality of modules assembled together, each module having a first and a second chamber or layer. The layers may be aligned and connected using oxygen plasma bonding. The holes at the inlets and outlets may be punched out and tubing connected to the inlets and outlets.
(34) It is therefore intended that the foregoing detailed description be regarded as illustrative rather than limiting, and that it be understood that it is the following claims, including all equivalents, that are intended to define the spirit and scope of this invention.

## Claims

1. A device for exchange between a first agent and a second agent; the device comprising: a first chamber having a first inlet and a first outlet with a first flow passageway extending between the first inlet and the first outlet for flowing a first agent through the first chamber; a second chamber having a second inlet and a second outlet with a second flow passageway extending between the second inlet and the second outlet for flowing a second agent through the second chamber, wherein the second chamber is a mirror image of the first chamber; and a membrane formed by a portion of walls of the first chamber and a portion of walls of the second chamber where the portion of the walls of the first chamber contact the portion of the walls of the second chamber along a central portion of the first chamber and a central portion of the second chamber and positioned between the first flow passageway and the second flow passageway to enable exchange through the membrane of the first agent with the second agent, wherein the first chamber and the second chamber each comprise flow distribution members that extend along a flow direction of the first agent and the second agent, respectively, and wherein the flow distribution members of each of the first chamber and the second chamber are tapered to enable uniform distribution of flow of the first agent through the first chamber and the second agent through the second chamber.
2. The device according to claim 1, wherein the first chamber and the second chamber each comprise a plurality of transport channels, wherein the plurality of transport channels form a pattern in another portion of the walls of the first chamber and the second chamber, wherein the pattern of the plurality of transport channels comprise a plurality of bifurcations, wherein a number of the plurality of bifurcations increases towards the central portions of the first chamber and the second chamber.
3. The device according to claim 1, wherein the first chamber comprises a first passageway and a second passageway and the second chamber comprises another first passageway and another second passageway.
4. The device according to claim 3, wherein the first passageway and the second passageway each comprise structural members formed in the first and second passageways.
5. The device according to claim 1, wherein the flow distribution members are positioned upstream from a plurality of transport channels in each of the first chamber and the second chamber.
6. The device according to claim 1, comprising a single module device.
7. The device according to claim 1, comprising a multiple module device comprising a plurality of the first chambers and a plurality of the second chambers.

8. The device according to claim 7, wherein a wall of the first chamber or a wall of the second chamber forms the membrane for each module of the multiple module device.
9. The device according to claim 1, wherein the first chamber and the second chamber comprise poly(dimethyl siloxane) (PDMS).
10. The device according to claim 1, wherein the membrane is permeable to a nitrosylating agent.
11. The device according to claim 10, wherein the nitrosylating agent comprising ethyl nitrite.
12. The device according to claim 2, wherein the plurality of transport channels of each of the first chamber and the second chamber are formed side-by-side such that the first agent and the second agent are flowed in a same flow direction.